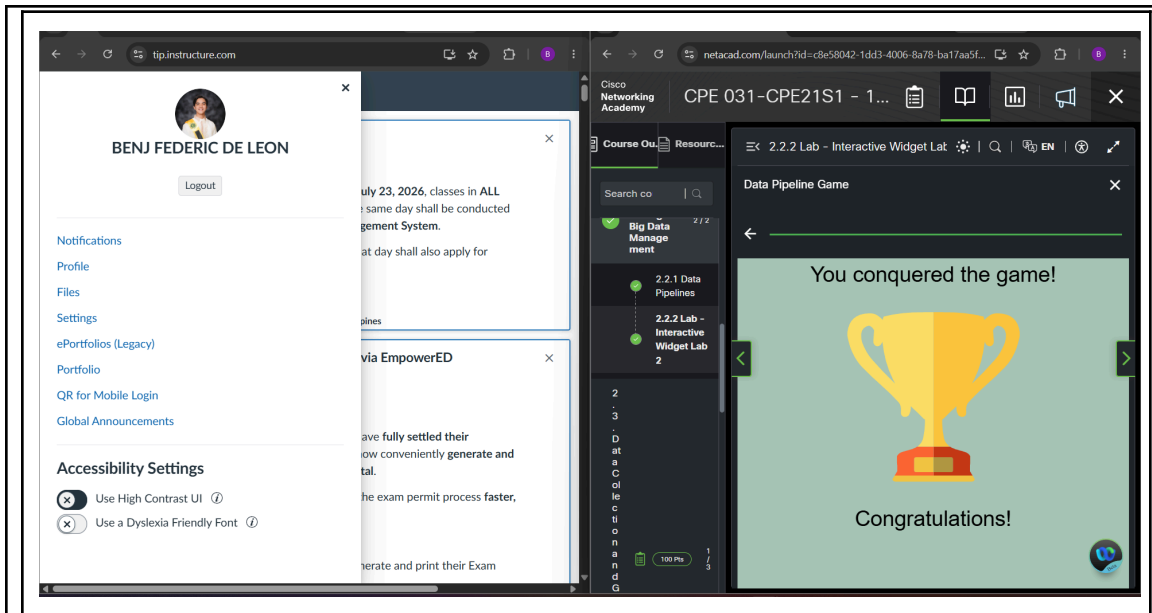


Activity No. 2 Interactive Widget	
Course Code: CPE 031	Program: BsCpE
Course Title: Visualizations and Data Analysis	Date Performed: Jul 25, 2026
Section: CPE21S1	Date Submitted: Aug 1, 2026
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1. Discussion	
This activity demonstrates the principles of Extract, Transform, Load(ETL) in a data engineering pipeline. It highlights the importance of data ingestion, data cleaning, and data transformation to ensure data quality before storing it in a destination database.	
2. Materials and Equipment	
Software: Interactive Data Pipeline Game (Lab 2.2.2: Interactive Widget Lab 2). Data Sources & Pipeline Stages: • Ingestion Nodes: Read JSON, Read XML, Read CVS. • Cleaning Nodes: Fix Corrupted Rows, Remove Duplicates, Delete Empty Rows. • Transformation Nodes: Standardize Case, Round Decimals, Standardize Units. • Storage Destination: Load into DB(Database target node).	
3. Procedure	
1. Data Ingestion (Read Data): • Selected Read CSV, which failed with a "Parsing Error: Not valid CSV". • Selected Read JSON, which successfully ingested the raw dataset into a structured table format. 2. Data Cleaning: • Attempted to load directly or standardize, triggering a "Data Error: Duplicate rows found". • Applied the Remove Duplicates processing stage to eliminate redundant user entries (e.g., duplicate rows for User 1). 3. Data Transformation: • Attempted to load data into the database, triggering a "Format Error: Inconsistent case" due to mixed text cases in the color column (e.g., RED, blue, BLUE). • Applied the Standardize Case transformation block to convert all text values in the color column to lowercase (red, blue, green). 4. Data Loading: • Connected the pipeline to the Load into DB node to successfully store the clean, standardized data in the target database.	
4. Output	



5. Conclusion/Analysis/Learnings

In this lab, I learned how messy raw data can be and why a good ETL pipeline is so important before saving anything to a database. Stepping through the game showed me that cleaning data isn't just one simple step, but a careful process, from picking the right tool to read JSON, to throwing out duplicate rows, and finally fixing mixed capital letters so everything looks the same. Catching and fixing those mistakes along the way made it clear that taking the time to check and clean data early on is what keeps a database working well and actually useful.

6. Assessment Rubric

